Sinais e Sistemas para Engenharia

MÓDULO 2A - VARIÁVEIS COMPLEXAS

VERSÃO OCTAVE

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Variáveis Complexas

Números Naturais $N = \{0,1,2,3\ \ldots.\}$

Números Inteiros $Z = \{\ldots - 3, -2, -1, 0, 1, 2, 3\ \ldots.\}$

Números Racionais $Q = \{x | x = {}^{a}/_{b}, com\ \ a \in Z, b \in Z\ e\ b \neq 0\}$

Números Reais $R = \{Q\ U\ I\}$

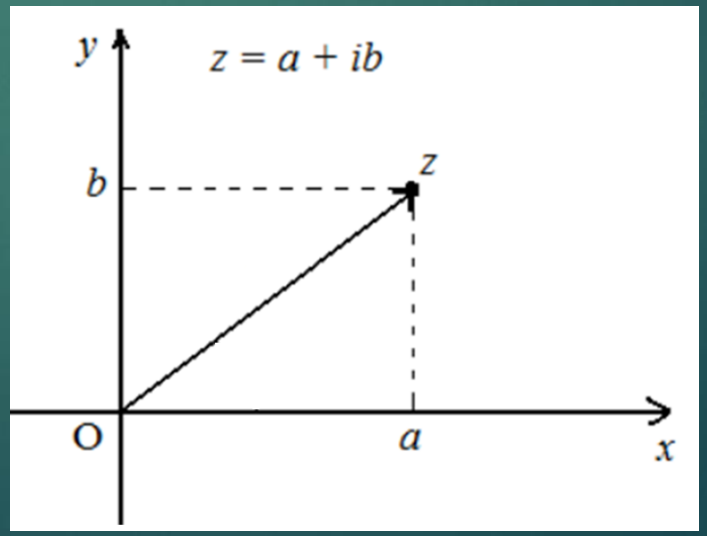
Números Complexos $C = \{R\ x\ R\}$

Números complexos são formados por um par ordenado de números reais (Vetor!).

$$\boldsymbol{z = a + jb},$$

$com\ \ a \in R, \quad b \in R \quad e \quad j = \sqrt{-1}$

$com \quad a = Re(z) \quad \text{e} \quad b = Im(z)$



Representação cartesiana

$z = a + jb$

$|z| = r = \sqrt{a^2 + b^2}$ - módulo de Z

$\theta = \text{atan}(^b/_a)\ ou\ \text{atan2}(b, a)$ - fase de Z

Obs. – $\text{atan}(^b/_a)$ não leva em conta o quadrante do ângulo de fase!

Representação polar/exponencial

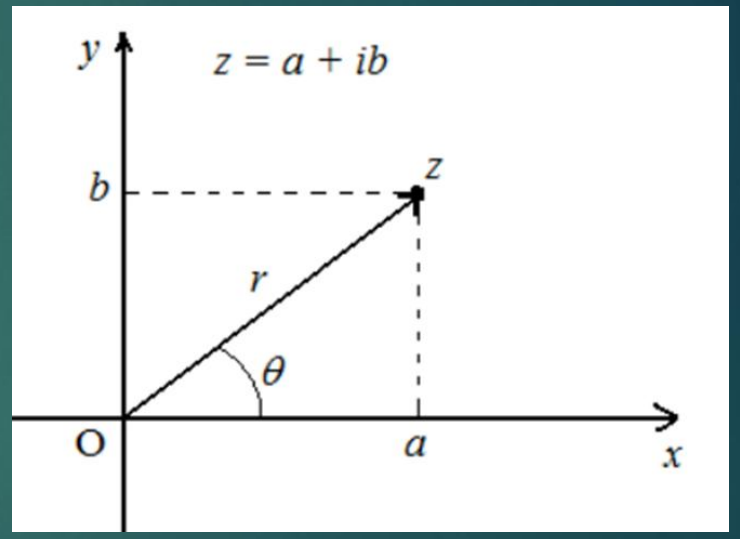
$z = re^{j\theta}$

a= $r\ \cos(\theta)$ parte real de z

b= $r\ \sin(\theta)$ parte imaginária de z

$$e^{j\theta} = \cos\theta + jsin(\theta)$$

Euler



$$z = a + jb = r\cos(\theta) + j\,r\sin(\theta) = re^{j\theta}$$

Exemplo

$$z_1 = 5 + j3 = (5\,,3) = 5.8309519e^{j0.5404195} = 5.83095\angle 30.9637^0$$

$$z_2 = 7 - j4 = (7,-4) = 8.0622577e^{-j0.5191461} = 8.0622577\angle -29.745^0$$

```
>> z1 = 5 +3*i
z1 =  5 + 3i
>>
>> z2 = 7-4*i
z2 =  7 - 4i
```

```
>> z1=complex(5,3)
z1 =  5 + 3i
>>
>> z2=complex(7,-4)
z2 =  7 - 4i
```

```
>> z1=5.8309519 * exp(i*0.5404195)
z1 =  5.0000 + 3.0000i
>>
>> z2=8.0622577 * exp(-i*0.5191461)
z2 =  7.0000 - 4.0000i
```

```
>> teta1 = 30.9637 * pi / 180;
>> z1=5.8309519 * exp(i*teta1)
z1 =  5.0000 + 3.0000i
>>
>> teta2 = 29.745 * pi / 180;
>> z2=8.0622577 * exp(-i*teta2)
z2 =  7.0000 - 4.0000i
```

```
function [zr, zi] = retangular(z)
    zr = real(z);
    zi = imag(z);
endfunction
```

```
function [z_m,z_f] = polar(z)
    z_m=abs(z);
    z_f=atan2(imag(z),real(z));
endfunction
```

```
>> [zr1 zi1] = retangular(z1)
zr1 = 5.0000
zi1 = 3.0000
>>
>> [zr2 zi2] = retangular(z2)
zr2 = 7.0000
zi2 = -4.0000
```

```
>> [zm1 zf1] = polar(z1)
zm1 = 5.8310
zf1 = 0.5404
>>
>> [zm2 zf2] = polar(z2)
zm2 = 8.0623
zf2 = -0.5191
```

Operações com números complexos

$z_1 = 5 + j3 = 5.831e^{j0.54042}$

$z_2 = 7 - j4 = 8.062e^{-j0.51915}$

Sejam os números complexos z_1 e z_2 definidos por:

$z_1 = a + jb = |z_1|e^{j\theta_1}$ $\qquad$ $z_2 = c + jd = |z_2|e^{j\theta_2}$

1) **Soma**

$z_1 + z_2 = (a+c) + j(b+d)$

$z_1 + z_2 = (5 + j3) + (7 - j4) = (5 + 7) + j(3 - 4) = 12 - j$

2) **Subtração**

$z_1 - z_2 = (a-c) + j(b-d)$

$z_1 - z_2 = (5 + j3) - (7 - j4) = (5 - 7) + j(3 - (-4)) = -2 + j7$

3) **Multiplicação**

Forma cartesiana:

$z_1 z_2 = (a + jb)(c + jd) = (ac-bd) + j(ad+bc)$

$z_1 z_2 = (5 + j3)(7 - j4) = 47 + i$

Forma polar

$z_1 z_2 = (a + jb)(c + jd) = |z_1|e^{j\theta_1}\,|z_2|e^{j\theta_2} = |z_1||z_2|e^{j(\theta_1+\theta_2)}$

$z_1 z_2 = 5.831e^{j0.54042}\ 8.062e^{-j0.51915} = (5.831)(8.062)e^{j(0.5404-0.51915)}47.01e^{j0.0212734}$

4) **Conjugado**

$z_1{}^* = a - jb$

$z_1{}^* = 5 - j3 = 5.831e^{-j0.54042}$

$z_2{}^* = 7 + j4 = 8.062e^{j0.51915}$

$z_1\, z_1{}^* = a^2 + b^2$ (real)

```
>> z1=complex(5,3)
z1 =   5 + 3i
>> z2=complex(7,-4)
z2 =   7 - 4i
>>
>> zr=z1+z2
zr =   12 -  1i
>>
>> zr=z1-z2
zr = -2 + 7i
>>
>> zr=z1*z2
zr =   47 +  1i
>>
>> zr=conj(z1)
zr =   5 - 3i
>> zr=conj(z2)
zr =   7 + 4i
>>
>> zr=z1*conj(z1)
zr = 34
```

$$z_1 = a + jb = |z_1|e^{j\theta_1} \qquad z_2 = c + jd = |z_2|e^{j\theta_2}$$

$$z_1 = 5 + j3 = 5.831e^{j0.54042}$$

$$z_2 = 7 - j4 = 8.062e^{-j0.51915}$$

5) Divisão

Forma cartesiana

$$\frac{z_1}{z_2} = \frac{z_1\, z_2^*}{z_2\, z_2^*} = \frac{(a+jb)}{(c+jd)}\frac{(c-jd)}{(c-jd)} = \frac{(\mathrm{ac+bd})+\mathrm{j}(\mathrm{bc-ad})}{\mathrm{c}^2+\mathrm{d}^2} = \frac{ac+bd}{c^2+d^2} + j\frac{bc-ad}{c^2+d^2}$$

$$\frac{z_1}{z_2} = \frac{(5+j3)}{(7-j4)} = \frac{23}{65} + j\left(\frac{41}{65}\right)$$

Forma polar

$$\frac{z_1}{z_2} = \frac{(a+jb)}{(c+jd)} = \frac{|z_1|e^{j\theta_1}}{|z_2|e^{j\theta_2}} = \frac{|z_1|}{|z_2|}e^{j(\theta_1-\theta_2)}$$

$$\frac{z_1}{z_2} = \frac{5.831e^{j0.54042}}{8.062e^{-j0.51915}} = \frac{5.831}{8.062}e^{j(0.54042+0.51915)} = 0.7232e^{j1.059567}$$

```
>> z1=complex(5,3)
z1 =  5 + 3i
>> z2=complex(7,-4)
z2 =  7 - 4i
>>
>> zr=z1/z2
zr =  0.3538 + 0.6308i
>>
>> [m,f]=polar(zr)
m = 0.7232
f = 1.0596
>>
>> zr=z1^2
zr =  16 + 30i
>>
>> [m,f]=polar(zr)
m = 34
f = 1.0808
```

6) Elevar ao quadrado – equivale a elevar o módulo ao quadrado e dobrar a fase

$$\mathrm{z}^2 = \begin{cases} \mathrm{r}^2 \\ 2\theta + 2\mathrm{k}\pi \end{cases}$$

$$z_1{}^2 = \left(5.831e^{j0.54042}\right)^2 = 5.831^2 e^{j2(0.54042)} = 34.0\, e^{j1.080834+2k\pi}$$

$$z_1 = a + jb = |z_1|e^{j\theta_1} \qquad z_2 = c + jd = |z_2|e^{j\theta_2}$$

$$z_1 = 5 + j3 = 5.831e^{j0.54042}$$

$$z_2 = 7 - j4 = 8.062e^{-j0.51915}$$

7) Elevar à potência n

$$z^n = \begin{cases} r^n \\ n\theta + 2k\pi \end{cases}$$

$$z_1{}^3 = \left(5.831e^{j0.54042}\right)^3 = 5.831^3 e^{j3(0.54042)} = 198.25236e^{j1.6212585+2k\pi}$$

8) Raiz quadrada – equivale a tirar a raiz quadrada do módulo e dividir o argumento por dois.

$$\sqrt{z} = \begin{cases} \sqrt{r} \\ \dfrac{\theta}{2} + \dfrac{2k\pi}{2} \end{cases}$$

$$\sqrt{z_2} = \sqrt{8.062e^{-j0.51915}} = \sqrt{8.0625}\, e^{-j\frac{0.51915}{2}} = 2.8394e^{-j(0.2595731+k\pi)}$$

$$-1 = -1 + j0 = 1.0\, e^{j\pi}$$

$$\sqrt{-1} = \sqrt{1.0\, e^{j\pi}} = \sqrt{1.0}\, e^{j\frac{\pi}{2}} = 0 + j = j$$

```
>> z1=complex(5,3)
z1 =  5 + 3i
>> z2=complex(7,-4)
z2 =  7 - 4i
>>
>> zr=z1^3
zr =  -10 + 198i
>>
>> [m,f]=polar(zr)
m = 198.25
f = 1.6213
>>
>> zr=sqrt(z2)
zr =  2.7443 - 0.7288i
>>
>> [m,f]=polar(zr)
m = 2.8394
f = -0.2596
>>
>> zr=sqrt(-1)
zr =  0 + 1i
>>
>> [m,f]=polar(zr)
m = 1
f = 1.5708
```

Números Complexos no plano cartesiano

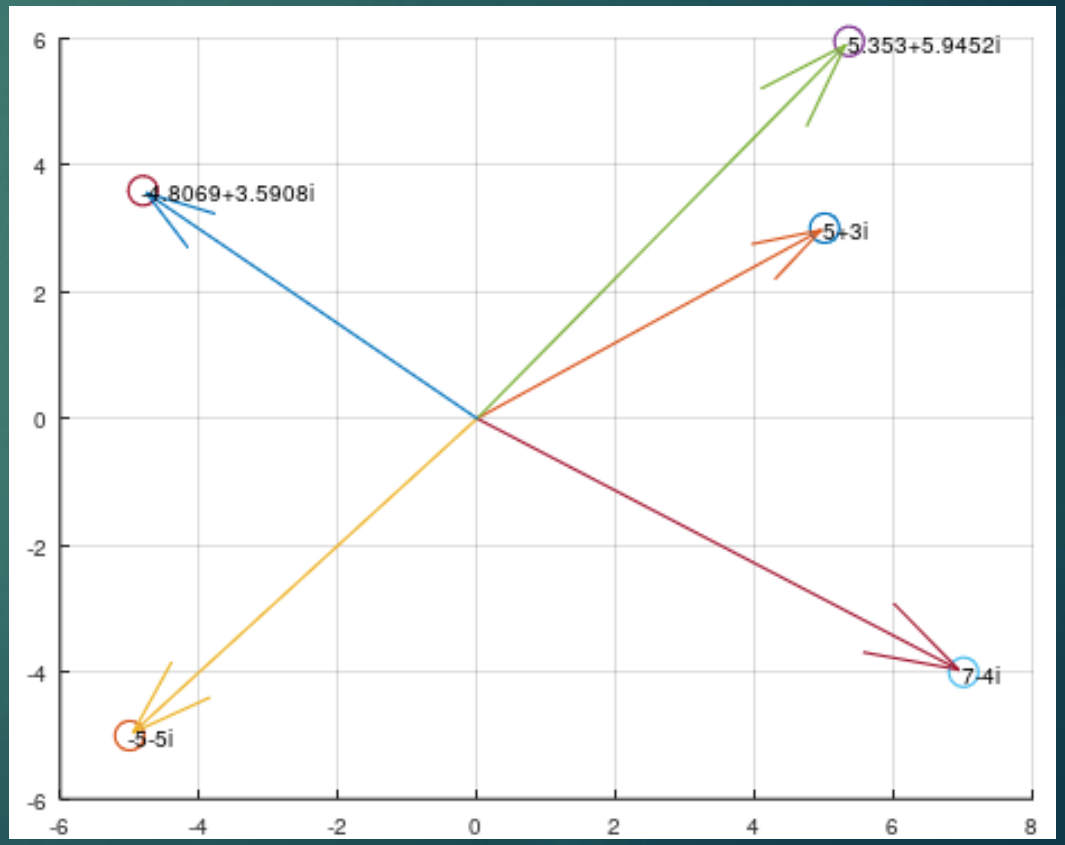
$z_1 = (5\,,3)$
$z_2 = (7\,,-4)$
$z_3 = 8\angle 48^0$
$z_4 = -5 - j5$
$z_5 = 6e^{j2.5}$

```
>> z(1)=complex(5,3);
>> z(2)=complex(7,-4);
>> z(3)=8*exp(i*48*pi/180);
>> z(4)=-5-5*i;
>> z(5)=6*exp(i*2.5);
```

```
>> z'
ans =

   5.0000 - 3.0000i
   7.0000 + 4.0000i
   5.3530 - 5.9452i
  -5.0000 + 5.0000i
  -4.8069 - 3.5908i
```

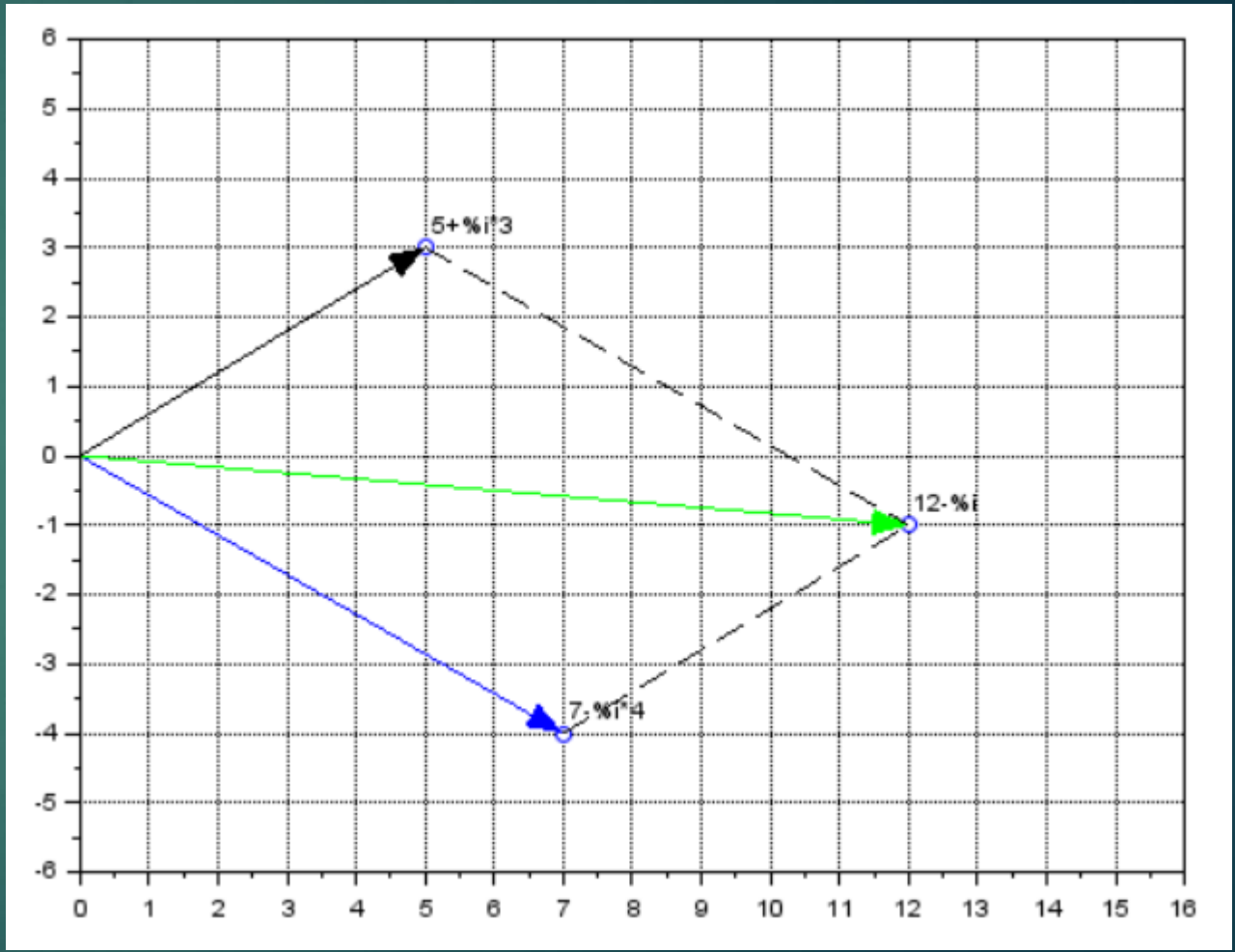
```
>> hold on; grid on;
>> for (k=1:length(z))
plot(real(z(k)),imag(z(k)),'o')
a=quiver(0, 0, real(z(k))*1.1, imag(z(k))*1.1);
text(real(z(k)),imag(z(k)),num2str(z(k)));
set(a,'maxheadsize',0.1);
end
```



Geometria da operações com números complexos

Soma $z = z_1 + z_2$

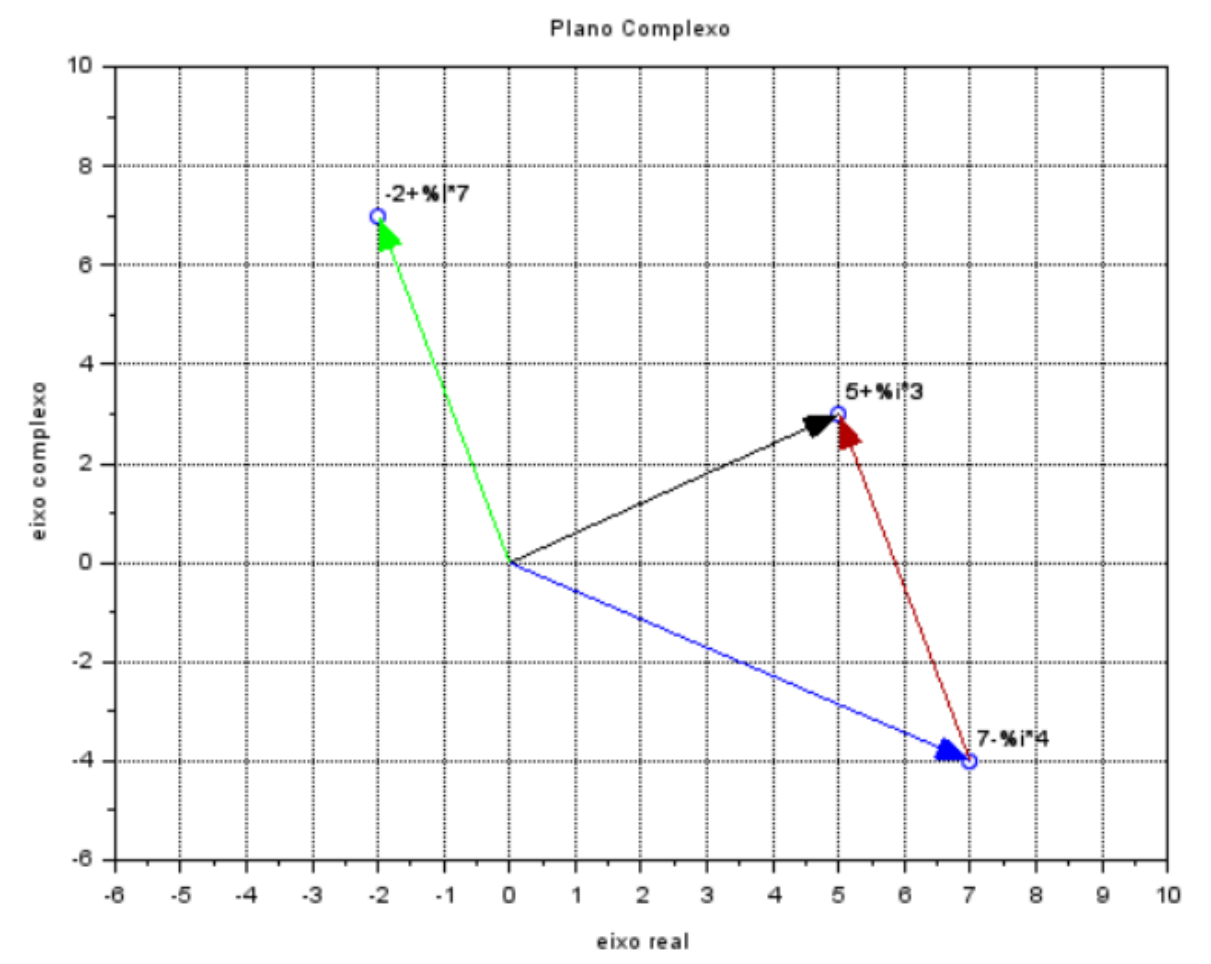
```
>> z1=complex(5,3)
z1 =  5 + 3i
>> z2=complex(7,-4)
z2 =  7 - 4i
>> zr=z1+z2
zr =  12 -  1i
```



Geometria da operações com números complexos

Subtração $z = z_1 - z_2$

```
>> z1=complex(5,3)
z1 =  5 + 3i
>> z2=complex(7,-4)
z2 =  7 - 4i
>> zr=z1-z2
zr = -2 + 7i
```



Geometria da operações com números complexos

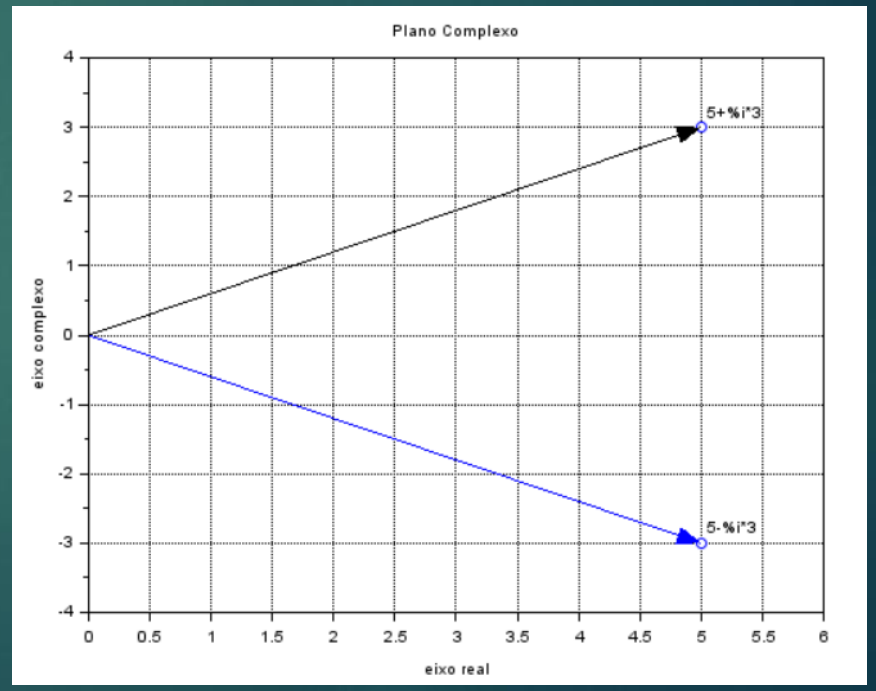
Conjugado $z = z_1^*$

```
>> z1=complex(5,3)
z1 =  5 + 3i
>> conj(z1)
ans =  5 - 3i
>> z1'
ans =  5 - 3i
>> conj(conj(z1))
ans =  5 + 3i
```

```
>> [zm1 zf1] = polar(z1)
zm1 = 5.8310
zf1 = 0.5404
>>
>> [zm1 zf1] = polar(conj(z1))
zm1 = 5.8310
zf1 = -0.5404
```

```
>> z=[z1;z1']
z =

   5 + 3i
   5 - 3i
```



Geometria das operações com números complexos

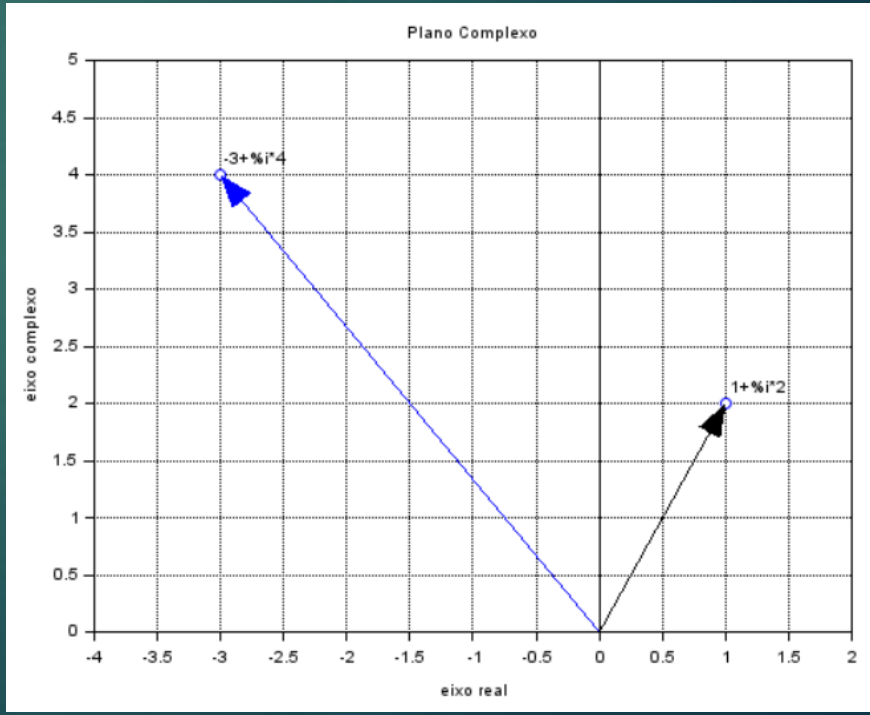
Elevar ao quadrado $z = z_1^2$

```
>> z1=complex(1,2)
z1 =  1 + 2i
>>
>> z1^2
ans = -3 + 4i
```

```
>> z1=complex(1,2)
z1 =  1 + 2i
>>
>> [zm1 zf1] = polar(z1)
zm1 = 2.2361
zf1 = 1.1071
>>
>> zm1^2 * exp(i* 2*zf1)
ans = -3.0000 + 4.0000i
```

```
>> z=[z1;z1^2]
z =

   1 + 2i
  -3 + 4i
```

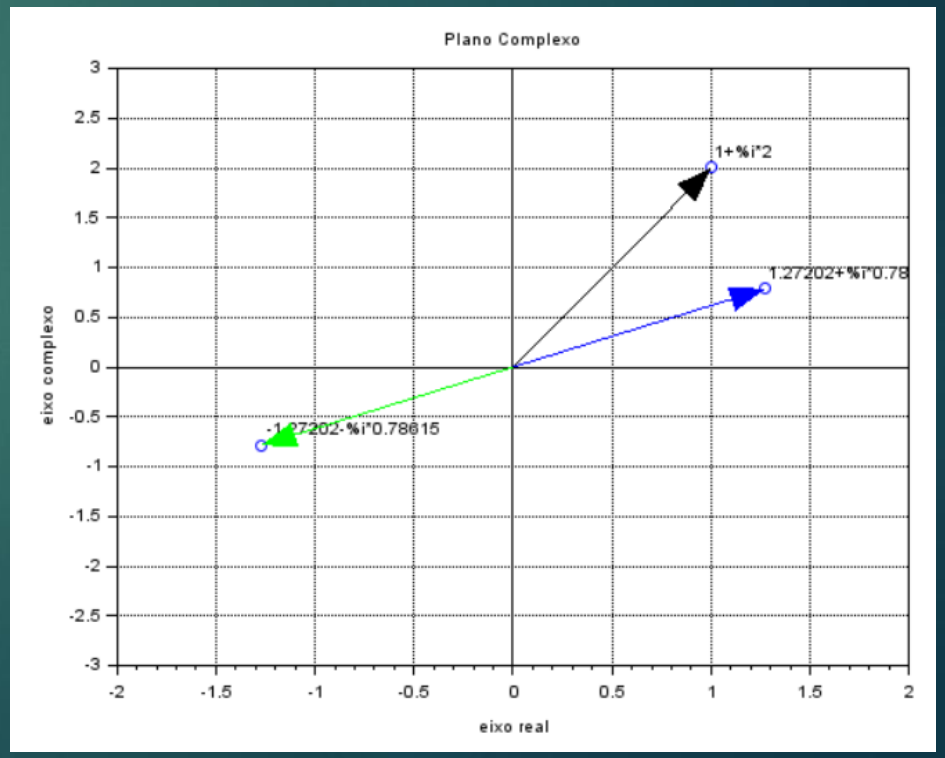


Geometria das operações com números complexos

Raiz quadrada $z = \sqrt{z_1}$

```
>> z1=complex(1,2)
z1 =  1 + 2i
>>
>> sqrt(z1)
ans =  1.2720 + 0.7862i
```

```
>> z1=complex(1,2)
z1 =  1 + 2i
>>
>> [zm1 zf1] = polar(z1)
zm1 = 2.2361
zf1 = 1.1071
>>
>> raiz1 = sqrt(zm1) * exp(i * zf1/2)
raiz1 =  1.2720 + 0.7862i
>>
>> raiz2 = sqrt(zm1) * exp(i * (zf1/2+2*pi/2))
raiz2 = -1.2720 - 0.7862i
```

```
>> raiz1^2
ans =  1 + 2i
>>
>> raiz2^2
ans =  1.0000 + 2.0000i
```

```
>> z = [z1; raiz1; raiz2]
z =

   1.0000 + 2.0000i
   1.2720 + 0.7862i
  -1.2720 - 0.7862i
```